

KYLE SHAFFER

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EDUCATION

Master of Science, Information Science May 2015
University of North Carolina at Chapel Hill
School of Data & Information Sciences
Thesis: Predicting Speech Acts in MOOC Forum Posts Using Conditional Random Fields
Advisor: Dr. Jaime Arguello

Bachelor of Arts, Philosophy May 2010
University of Wisconsin - Eau Claire, *Magna Cum Laude*

RESEARCH INTERESTS

Sociology of Technology, Sociology of Knowledge, Epistemic Fragmentation, Technology & Society, Ethics of AI

ACADEMIC RESEARCH EXPERIENCE

Research Assistant August 2014 - May 2014
University of North Carolina at Chapel Hill *School of Data & Information Sciences*

- Research assistant to Dr. Jaime Arguello developing machine learning models for automatically predicting student information needs in online course discussion forums.
- Cleaned, analyzed and formatted data for machine learning experiments, trained and evaluated models, and contributed literature review and methods section of paper.
- Findings presented in paper "Predicting Speech Acts in MOOC Forum Posts" accepted to International Conference on Web and Social Media 2015.

Research Assistant August 2014 - May 2014
University of North Carolina at Chapel Hill *School of Medicine*

- Research assistant to Dr. Carlton Moore to investigating natural language processing techniques for medical record abstraction.
- Findings presented at poster session in American Medical Informatics Association 2015 Symposium.

Data Science Fellow June 2014 - August 2014
Georgia Institute of Technology *Data Science for Social Good Fellowship*

- Fully funded fellowship for data science students for applied research projects focused on positive community outcomes for the Atlanta metro area.
- Built interactive data visualization dashboard for insights into crowd-sourced data on cycling conditions in Atlanta metro area collected through [Cycle Atlanta](#) mobile app.
- Designed web-app using combinations of frontend technologies including HTML, CSS, JavaScript, and D3.js visualization library.
- Cleaned, ingested, and analyzed data collected through smartphones for monitoring rides throughout the city of Atlanta.

Undergraduate Research Assistant January 2014 - May 2014
University of Wisconsin-Eau Claire *Department of Linguistics*

- Research assistant to Dr. Lynsey Wolter investigating Zeno Vendler's semantics and philosophy of language.
- Conducted extensive literature review and contributed paper investigating the intersection between Vendler's semantics and analytic philosophy of language.

- Findings presented at poster session in “Zeno Vendler and Semantics of Believe and Know.”

PUBLICATIONS

Melissa Roemmele, **Kyle Shaffer**, Katrina R. Olsen, Yiyi Wang and Steve DeNeefe

AbLit: A Resource for Analyzing and Generating Abridged Versions of English Literature. *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL '23)*, 2023.

Kyle Shaffer

Language Clustering for Multilingual Named Entity Recognition. *Findings of the Association for Computational Linguistics: EMNLP 2021*.

Nathan O. Hodas, **Kyle Shaffer**, Artem Yankov, Courtney D. Corley Beyond Fine Tuning: Adding Capacity to Leverage Few Labels. *31st Conference on Neural Information Processing Systems, Limited Labeled Data Workshop*. 2017.

Svitlana Volkova, **Kyle Shaffer**, Jin Yea Jang, Nathan O. Hodas

Separating Facts from Fiction: Linguistic Models to Classify Suspicious and Trusted News Posts on Twitter. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL '17)*, 2017. (18% short paper acceptance rate)

Lawrence Phillips, **Kyle Shaffer**, Dustin Arendt, Nathan O. Hodas, Svitlana Volkova

Intrinsic and Extrinsic Evaluation of Spatiotemporal Text Representations in Twitter Streams. *Proceedings of the 2nd Workshop on Representation Learning for NLP*. 2017.

Jaime Arguello and **Kyle Shaffer**

Predicting Speech Acts in MOOC Forum Posts. In *Proceedings of the 9th International Conference on Web and Social Media (ICWSM '15)*, 2015. (19% acceptance rate)

Carlton Moore, **Kyle Shaffer**, Anna Kucharska-Newton, Stephanie Haas, Gerardo Heiss

Using Natural Language Processing to Facilitate Medical Record Abstraction in Epidemiological Studies. Poster presented at: *American Medical Informatics Association (AMIA) 2015 Annual Symposium*, 2015.

Kyle Shaffer

The Skeptic’s Language Game: Does Sextus Empiricus Violate Normal Language-Use? *Stance: An International Undergraduate Philosophy Journal*, Vol. 3, 2010.

INDUSTRY RESEARCH EXPERIENCE

Research Scientist, Natural Language Processing

Sep 2020 – May 2026

Language Weaver (RWS Group)

- Conducted applied research in natural language processing and machine learning in areas such as document classification, named entity recognition, and text generation.
- Published findings in top natural language processing conference proceedings.
- Analyzed, cleaned, and standardized large-scale datasets for use in machine learning projects across the broader applied research team.

Senior Data Scientist

Dec 2018 – Sep 2020

Qntfy

- Trained and evaluated natural language processing models for investigating social media phenomena for research projects funded by DARPA and the Department of Defense.
- Implemented advanced deep learning techniques from research literature including transfer learning, multi-task learning, and multi-modal modeling.

Data & Applied Scientist II

Jun 2016 – May 2017

Pacific Northwest National Laboratory

- Developed and validated deep learning models for application areas in language and image analysis for federally funded research projects relevant to state and national security.
- Cleaned, standardized, and analyzed large-scale social media datasets for use in research at the intersection of machine learning and computational social science.
- Performed in-depth literature reviews and research of latest state-of-the-art methods and contributed to writing research papers for conference publications.

TEACHING EXPERIENCE

Graduate Level Philosophy Tutor

Marquette University

November 2010 - December 2010

Tutoring Center

- Tutored graduate course in Epistemology for non-Philosophy grad students.
- Supplemented course lectures, assisted in understanding and interpretation of texts and philosophical concepts.
- Helped develop paper topics and provided feedback on drafts.

Philosophy Tutor

Marquette University

August 2010 - October 2010

Tutoring Center

- Tutored undergraduate course in Logic & Critical Reasoning.
- Provided feedback on paper proposals and drafts.
- Supplemented lectures with discussion sessions.

Undergraduate Teaching Assistant

University of Wisconsin-Eau Claire

January 2010 - May 2010

Department of Psychology

- Served as undergraduate teaching assistant for course in Cognitive Psychology for Dr. Allen Keniston.
- Graded student papers and provided feedback on paper proposals.

GRANTS, HONORS, AND AWARDS

Data Science For Social Good Fellowship

Georgia Institute of Technology

June 2014

- Fully funded fellowship for data science students for applied research projects focused on positive community outcomes for the Atlanta metro area.

Student Research Travel Grant

University of Wisconsin-Eau Claire

April 2010

Department of Philosophy & Religious Studies

- Student grant to offset travel costs of presenting undergraduate research at conference.
- Received for presentation of paper at 5th Annual Undergraduate Conference in Philosophy, SUNY-Oneonta, April 15-17, 2010.

Student-Faculty Research Grant

University of Wisconsin-Eau Claire

January 2010

Department of Linguistics

- Research grant awarded to exceptional student-faculty research projects.
- Received for research that would lead to presentation of research paper "Zeno Vendler and the Semantics of *Know* and *Believe*".